

LMA Phase 2 Project Submission

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Languages

High-Resource: Telugu | Low-Resource: Nepali

Transformer Architecture

Both the languages follow a similar transformer architecture here are the specifics about each module

Token and Positional Embeddings

Token embeddings map each token to a learned 384-dimensional vector, while learned positional embeddings provide position information for a maximum sequence length of 512. The two embeddings are added element-wise and embedding dropout with a rate of 0.1 is then applied:

$$x = E_{\text{token}} + E_{\text{position}}, \quad x = \text{Dropout}(x).$$

The learned positional embedding is simple to implement, but it limits the input length to 512 tokens. Therefore, longer texts must be divided into smaller chunks before being provided to the model.

Causal Masking

A causal mask is applied to the attention scores to prevent each token from attending to future tokens. The upper-triangular positions of the mask are set to $-\infty$, while the current and previous positions remain unmasked. After applying the mask, the softmax assigns zero attention probability to future positions, ensuring that predictions depend only on the current and previous tokens.

Empirical Verification

To verify causal masking, the future token at position $t+1$ was changed while keeping all tokens up to position t unchanged. For the sequences [2257, 5729, 2495, 1370, 3807] and [2257, 5729, 2495, 1370, 3808] the logits at $t=3$ were compared. The maximum absolute difference was 0.0, confirming that changing a future token does not affect the logits at the current position.

Transformer Block

- **Multi-head causal self-attention:** Each block applies multi-head self-attention using separate query, key, and value projections, with a causal mask to prevent attention to future tokens.
- **Feed-forward network:** A two-layer position-wise feed-forward network is used with GELU activation, expanding the dimension from $d_{\text{model}} = 384$ to $d_{\text{ff}} = 1536$ before projecting it back to 384.
- **Residual connections:** Residual connections are applied around both the attention and feed-forward sublayers, allowing the input of each sublayer to be added back to its output.
- **Layer normalization:** Pre-normalization is used, where LayerNorm is applied before both the attention and feed-forward sublayers. This provides more stable training for deeper Transformer stacks.
- **Dropout:** Dropout with a rate of 0.1 is applied to the attention probabilities and to the outputs of both the attention and feed-forward sublayers.

Twelve such Transformer blocks are stacked sequentially, followed by a final LayerNorm before the output projection to the vocabulary.

Telugu Phase 2 model Configurations

The configuration used for the Telugu GPT-style Transformer is shown below:

Parameter	Value
Vocabulary Size (V)	10,000
Maximum Sequence Length	512
Model Dimension (d_{model})	384
Number of Transformer Layers	12
Number of Attention Heads	8
Feed-Forward Dimension (d_{ff})	1,536
QKV Bias	False
Dropout Rate	0.1

Table 1: Telugu Phase 2 Transformer configuration

$$\text{Total Parameters} = 3,840,000 + 196,608 + (12 \times 1,773,312) + 768 = \boxed{25,317,120}$$

$$\boxed{\text{Total Trainable Parameters} \approx 25.32 \text{ Million}}$$

Telugu Training Configurations

The training configuration used for the Telugu GPT-style Transformer is shown below.

Parameter	Value
Batch Size	32
Gradient Accumulation Steps	1
Maximum Learning Rate	6×10^{-4}
Minimum Learning Rate	6×10^{-5}
Warmup Steps	2,000
Weight Decay	0.01
Adam β Values	(0.9, 0.95)
Adam ϵ	1×10^{-8}
Gradient Clipping	1.0
Logging Interval	50 steps
Evaluation Interval	1,000 steps
Evaluation Batches	40
Checkpoint Save Interval	1,000 steps
Archive Interval	5,000 steps
Target Training Tokens	550 million
PyTorch Compile	Enabled
Precision	precision
Maximum Consecutive Non-finite Skips	20
AMP Resume Scale Cap	65,536

Analysis of Telugu Model

Loss Curves

The training loss decreases steadily from approximately 9.1 to 4.2, indicating that the model is progressively learning the training data. The validation loss also decreases initially and gradually stabilizes around 5.0, showing continued improvement in generalization with no significant increase in validation loss.

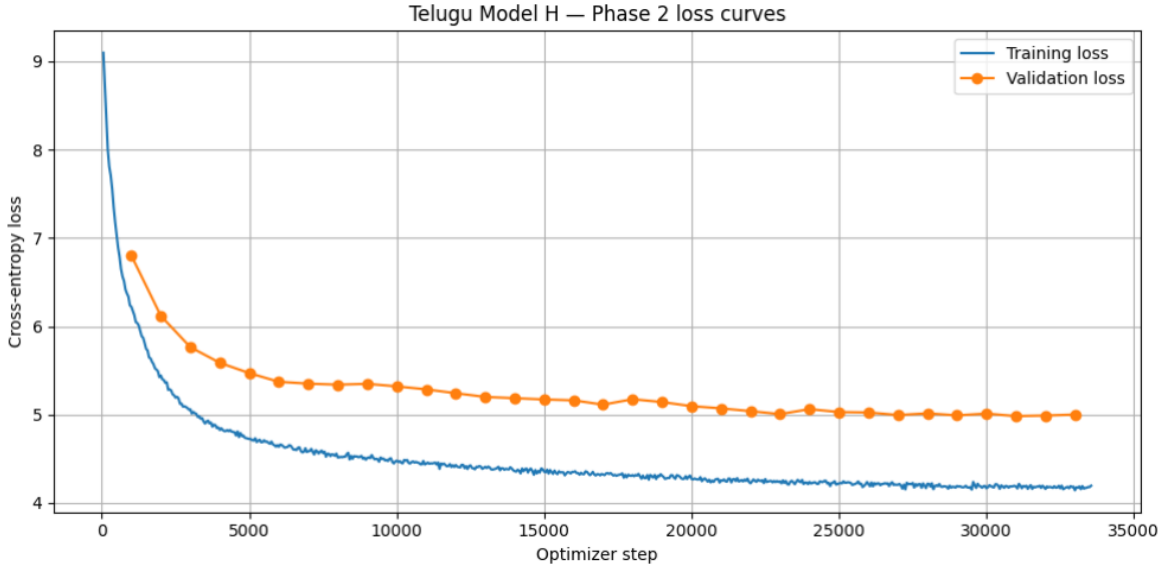


Figure 1: Training and validation loss curves for the Telugu Model H during Phase 2 training.

Validation Metrics

The final validation cross-entropy loss is 5.020426, corresponding to a perplexity of 151.475794. The model achieves 0.660157 bits-per-byte (BPB) over approximately 1.64 million evaluated tokens, indicating the model’s predictive performance on the held-out validation data.

Metric	Value
Cross-Entropy	5.020426
Perplexity	151.475794
Bits-per-Byte (BPB)	0.660157
Evaluated Tokens	1,638,400
Evaluated UTF-8 Bytes	17,975,776

Generation Evaluation

The generation results for different decoding strategies are shown below:

Strategy	BLEU-4	chrF	ROUGE-L	Distinct-1	Distinct-2	Repeated-3gram-
Greedy	0.584938	10.985178	11.182346	0.096680	0.131944	0.847782
Temperature 0.5	0.807188	17.963811	10.067168	0.323242	0.558532	0.328629
Temperature 1.0	0.667935	22.400999	9.223221	0.654297	0.967262	0.008065
Temperature 1.5	0.341415	23.185740	3.659606	0.826172	0.999008	0.000000

Table 2: Generation evaluation across different decoding strategies

As the temperature increases, the generated text becomes increasingly diverse, as shown by the rise in Distinct-1 and Distinct-2. The repeated 3-gram rate decreases substantially, from 84.78% with greedy decoding to 0% at temperature 1.5, indicating reduced repetition. Temperature 0.5 achieves the highest BLEU-4 score of 0.8072, showing the highest similarity to the reference text among the tested strategies. Temperature 1.0 provides a better balance between similarity and diversity, with high Distinct-2 (0.9673) and very low repetition (0.81%). At temperature 1.5, diversity is highest but BLEU-4 and ROUGE-L decrease, indicating a trade-off between diversity and reference similarity.

Attention Entropy Analysis

The mean attention entropy decreases from the early layers and reaches its minimum around layer 6, indicating that the attention becomes more focused in the middle layers. It then increases in the later layers, suggesting that attention becomes more distributed across tokens.

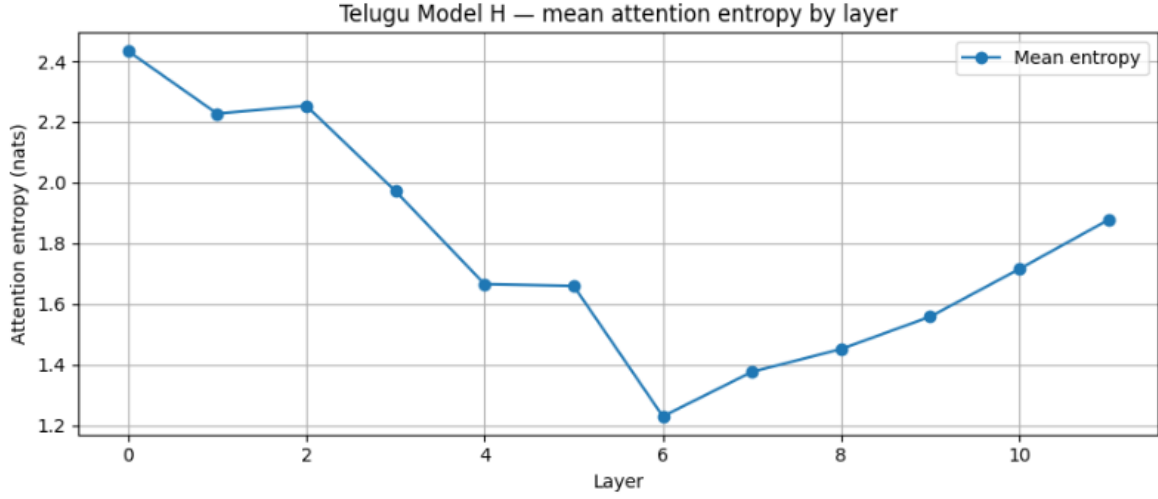


Figure 2: Mean attention entropy across layers of Telugu Model H.

Attention Distance Analysis

The mean query-key attention distance decreases significantly from the early layers to layer 6, indicating a stronger focus on nearby tokens. In the later layers, the distance increases, suggesting that the model increasingly captures longer-range relationships.

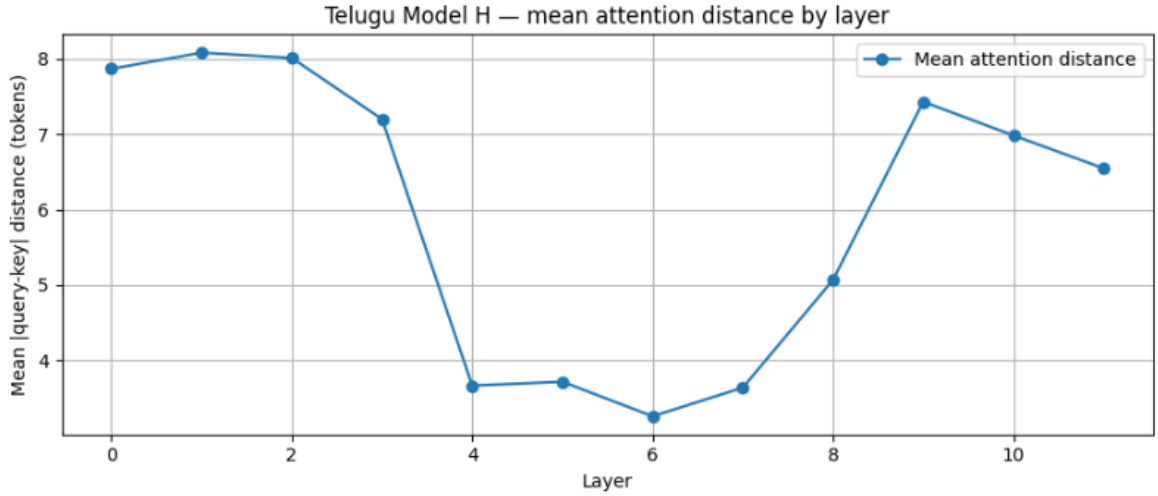


Figure 3: Mean attention distance across layers of Telugu Model H.

Nepali Model Analysis

The Nepali GPT-style Transformer follows the same overall decoder-only Transformer architecture as the Telugu model, with a model dimension of 384, 12 Transformer layers, 8 attention heads, and a feed-forward dimension of 1536. The model uses learned absolute positional embeddings, pre-normalization, and tied input-output embeddings.

Nepali Model Configuration

Parameter	Value
Vocabulary Size (V)	10,000
Maximum Sequence Length	512
Model Dimension (d_{model})	384
Number of Transformer Layers	12
Number of Attention Heads	8
Feed-Forward Dimension (d_{ff})	1,536
QKV Bias	False
Dropout Rate	0.1
Position Encoding	Learned Absolute
Normalization	Pre-Norm
Tied Embeddings	True

Nepali Training Configuration

The Nepali model was trained using a batch size of 32 with gradient accumulation over 2 steps. AdamW optimization was used with a maximum learning rate of 5×10^{-5} , a minimum learning rate of 5×10^{-6} , and 1,000 warmup steps.

Parameter	Value
Random Seed	1337
Batch Size	32
Gradient Accumulation Steps	2
Maximum Learning Rate	5×10^{-5}
Minimum Learning Rate	5×10^{-6}
Warmup Steps	1,000
Weight Decay	0.1
Adam β Values	(0.9, 0.95)
Adam ϵ	1×10^{-8}
Gradient Clipping	1.0
Logging Interval	100 steps
Evaluation Interval	500 steps
Evaluation Batches	50
Checkpoint Save Interval	1,000 steps
Target Training Tokens	550 million

Analysis of Model and Nepali

Loss Curves

The training loss decreases steadily from approximately 9.0 to 4.7, while the validation loss decreases from approximately 7.6 to 5.3. The continued decrease in both curves indicates that the model is learning progressively, although a gap between training and validation loss remains towards the end.

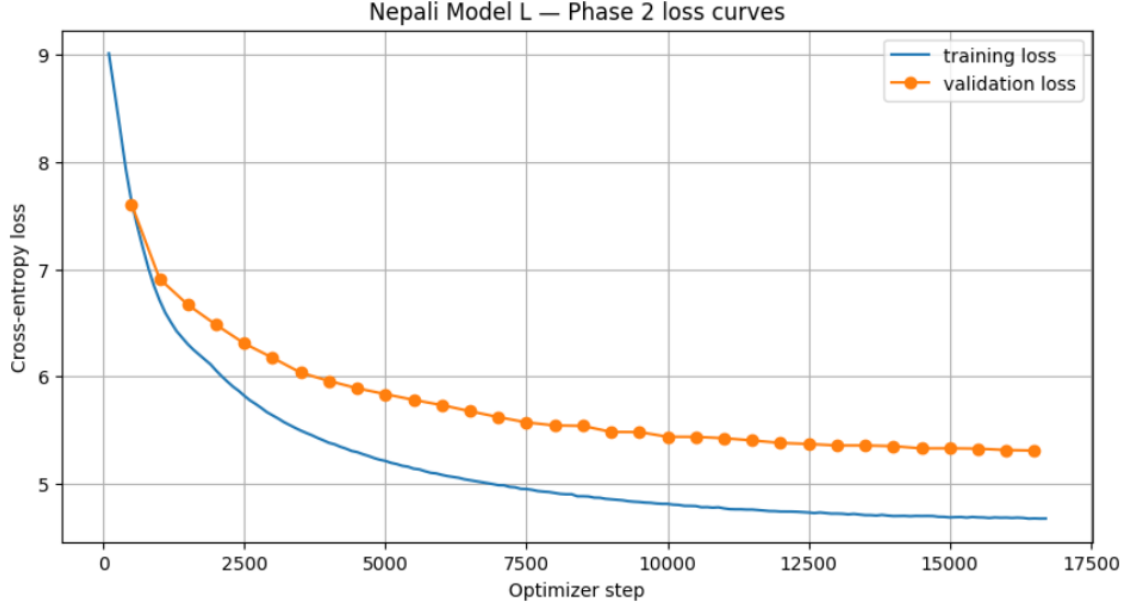


Figure 4: Training and validation loss curves for the Nepali Model L during Phase 2 training.

Generation Evaluation

The generation results for different decoding strategies are shown below:

Strategy	BLEU-4	chrF	ROUGE-L	Distinct-1	Distinct-2	Repeated-3gram-
Greedy	1.015664	15.830968	11.092134	0.157227	0.297619	0.621976
Temperature 0.5	0.964038	20.982782	11.057505	0.390625	0.752976	0.098790
Temperature 1.0	0.720886	20.941415	8.124324	0.678711	0.965278	0.002016
Temperature 1.5	0.445161	21.819186	3.119236	0.846680	1.000000	0.000000

Table 3: Generation evaluation across different decoding strategies for Nepali Model L

As the temperature increases, the generated text becomes more diverse, with Distinct-1 and Distinct-2 increasing substantially. The repeated 3-gram rate decreases from 62.20% with greedy decoding to 0% at temperature 1.5, showing reduced repetition. Greedy decoding gives the highest BLEU-4 score of 1.015664, while temperature 1.0 provides a strong balance between diversity and low repetition. At temperature 1.5, diversity is highest, but BLEU-4 and ROUGE-L decrease, showing the trade-off between diversity and reference similarity.

Attention Distance Analysis

The mean attention distance is high in the early layers and decreases significantly towards layer 6, reaching its minimum of approximately 3.2 tokens. It then increases in the later layers, indicating that the middle layers focus more on nearby tokens while deeper layers capture longer-range relationships.

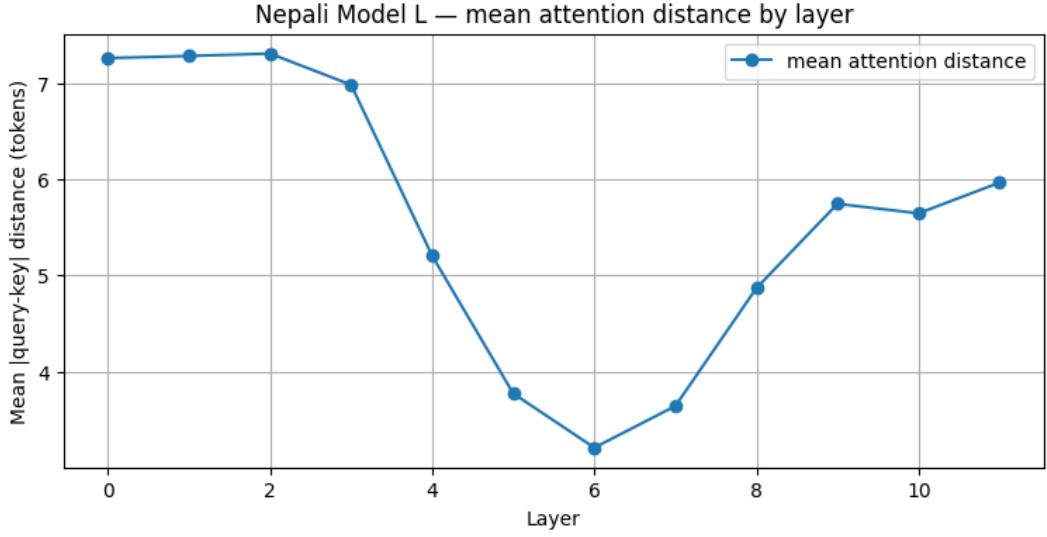


Figure 5: Mean attention distance across layers of Nepali Model L.

Attention Entropy Analysis

The mean attention entropy decreases from approximately 2.46 in the first layer to a minimum of about 1.70 around layer 7, indicating increasingly focused attention in the middle layers. It subsequently increases in the later layers, suggesting that attention becomes more distributed across tokens.

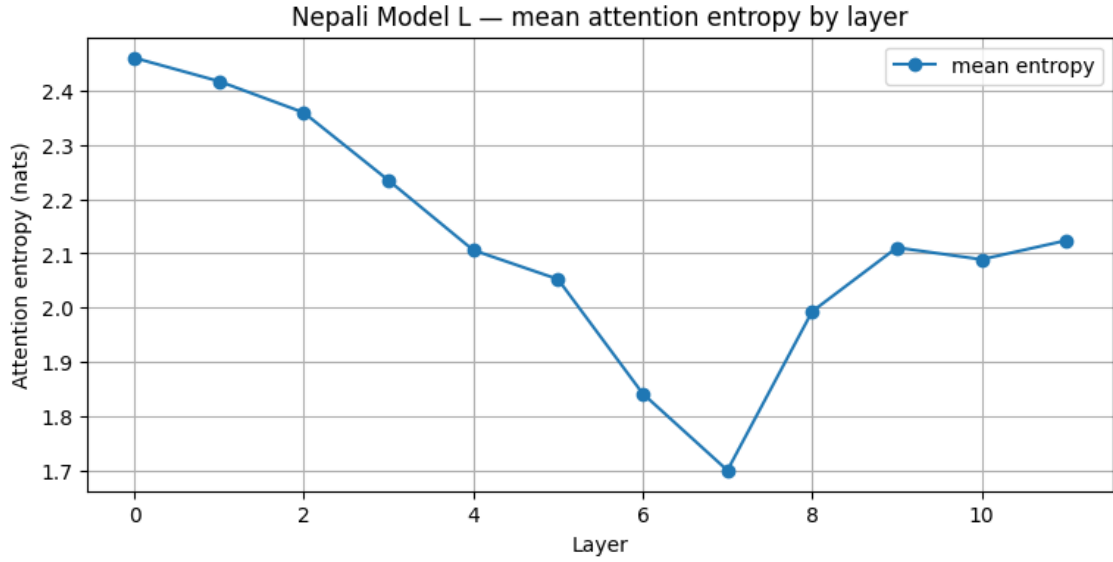


Figure 6: Mean attention entropy across layers of Nepali Model L.